INDUSTRIAL ROBOTICS PROJECT

Autonomous Drone Surveillance Mission

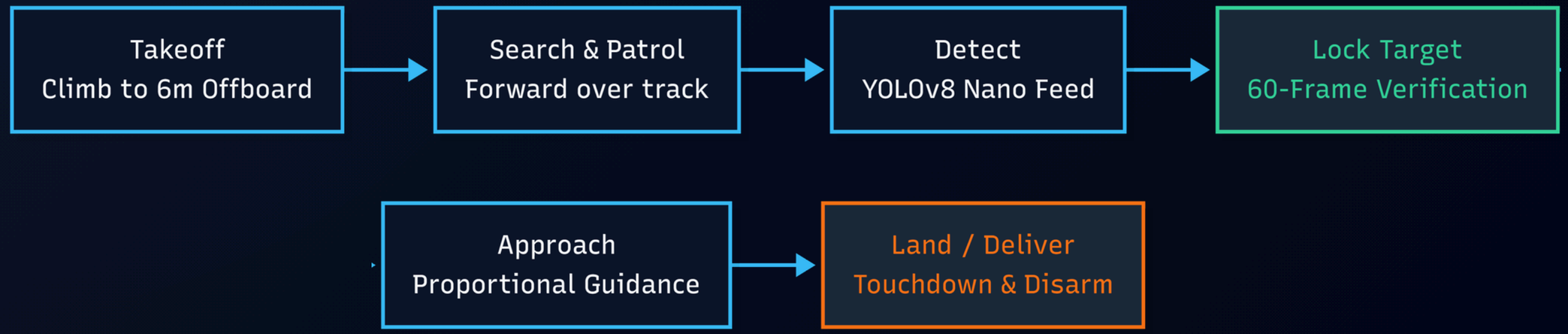
Advanced system leveraging PX4, ROS 2, and YOLOv8 for perception-to-action autonomous flight simulation.

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Mission Objective

The Challenge

Developing a **fully autonomous** UAV loop:

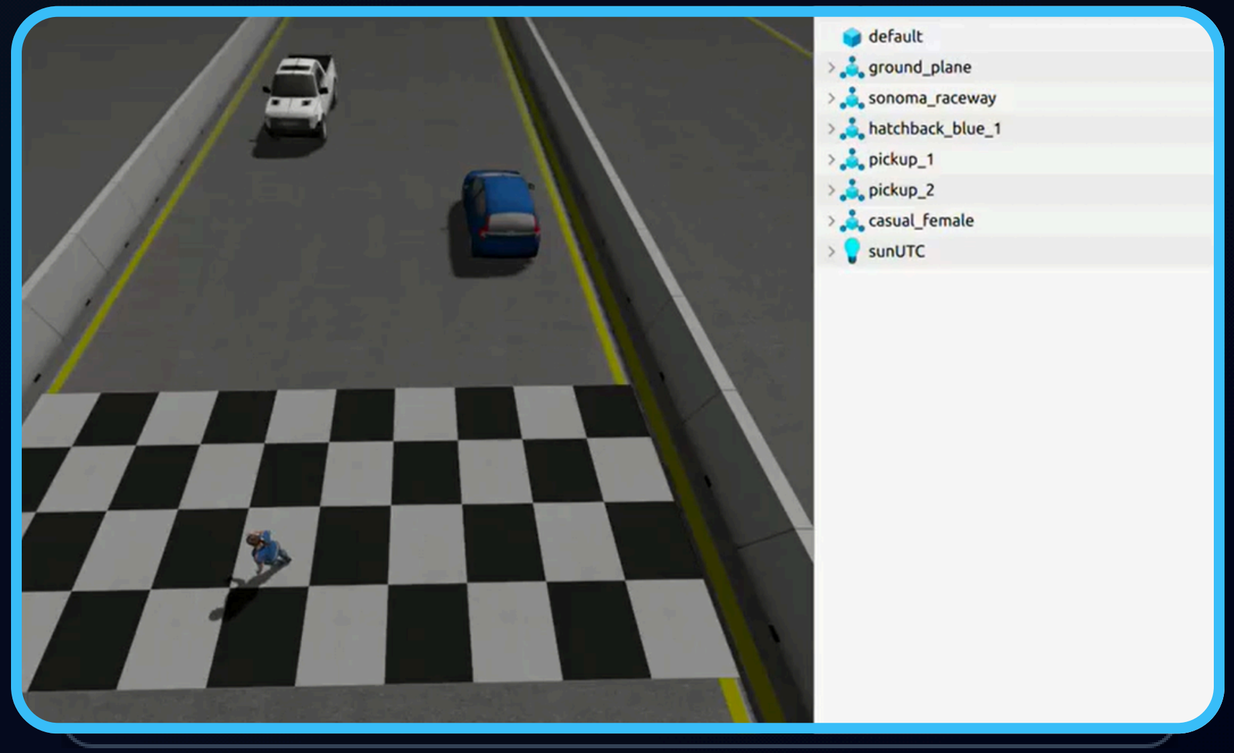


Mission Objective

The Challenge

Developing a **fully autonomous** UAV loop:

- **PX4 SITL:** Software-In-The-Loop simulation.
- **Computer Vision:** Real-time YOLOv8 detection.
- **ROS 2 Humble:** High-performance communication.



End-to-End Architecture

Perception

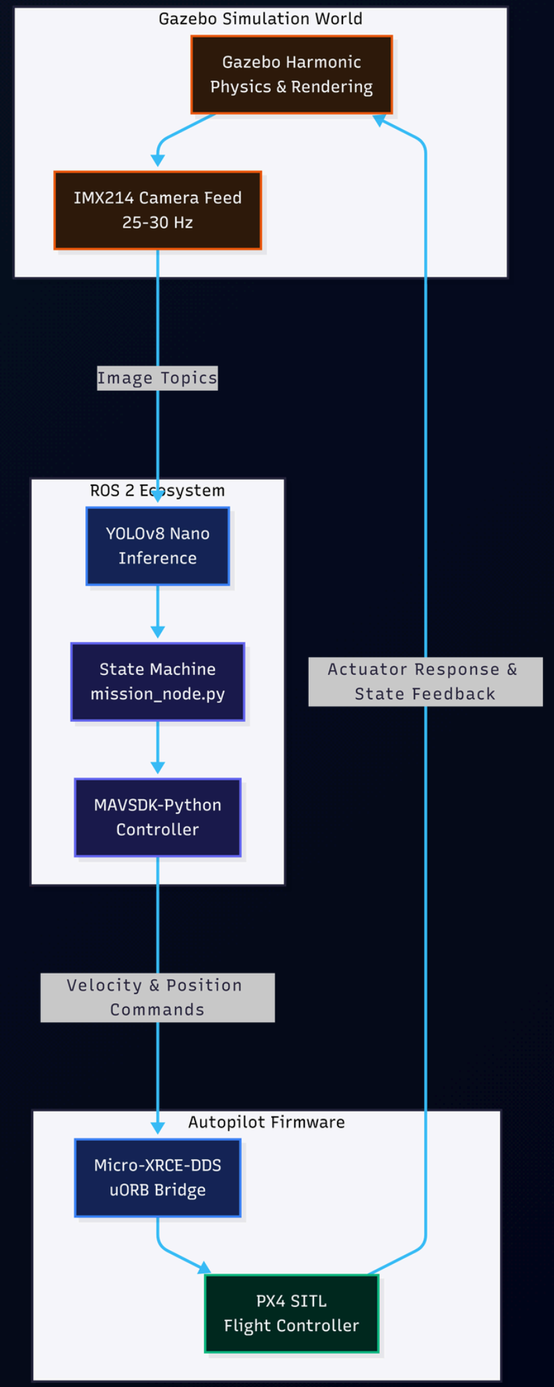
Bridges Gazebo camera topics to ROS 2 for YOLOv8 inference. Identifies target coordinates in image frame.

Decision

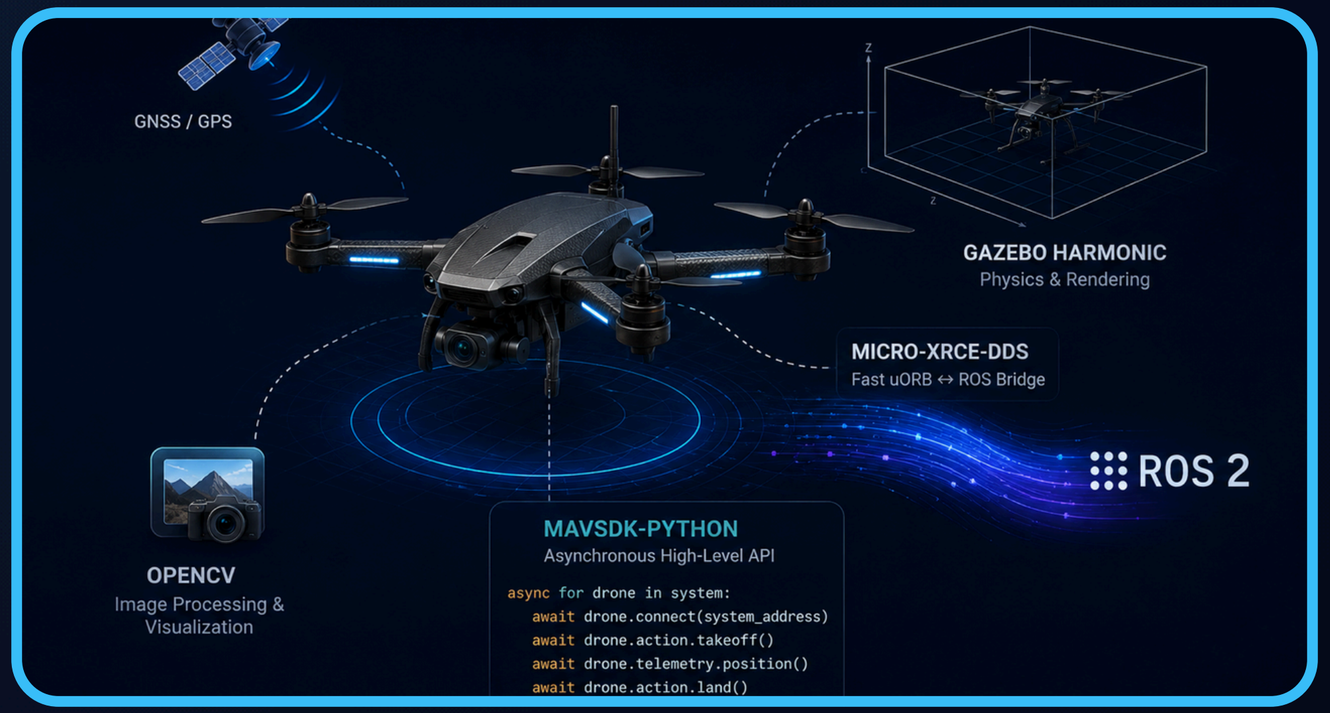
Asynchronous state machine (mission_node.py) manages navigation, lock-filters, and guidance logic.

Control

MAVSDK commands the PX4 autopilot using offboard velocity and position setpoints.



Software Integration



Our architecture utilizes **DDS (Data Distribution Service)** for seamless integration between the autopilot and the user logic.

- **Gazebo Harmonic:** Physics & Rendering.
- **Micro-XRCE-DDS:** Fast uORB ↔ ROS bridge.
- **MAVSDK-Python:** Asynchronous high-level API.
- **OpenCV:** Image processing & visualization.

PHASE 01

Gazebo Setup & Model Calibration

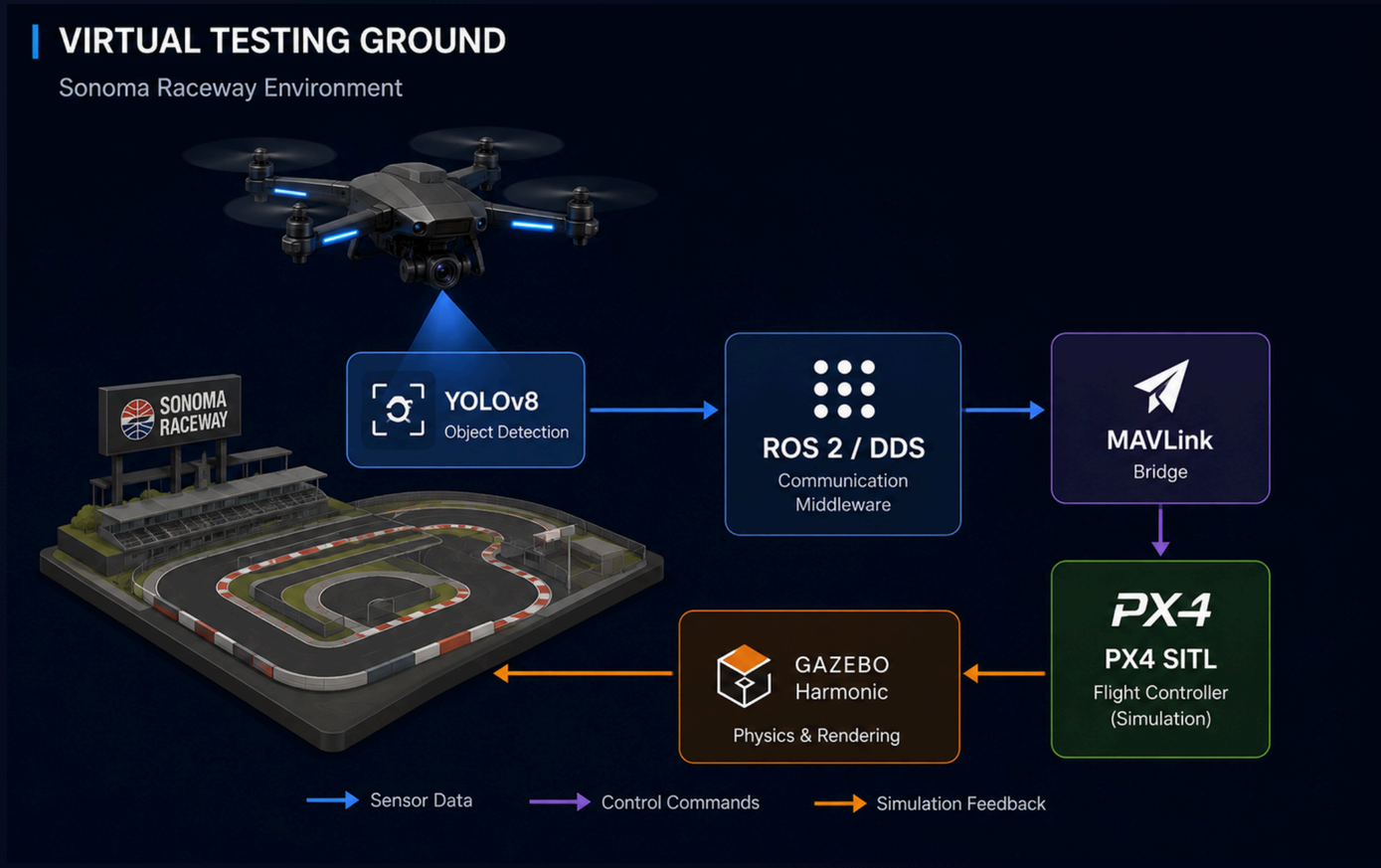
Establishing the high-fidelity racetrack environment and the x500 drone model with customized sensor suites.

Virtual Testing Ground

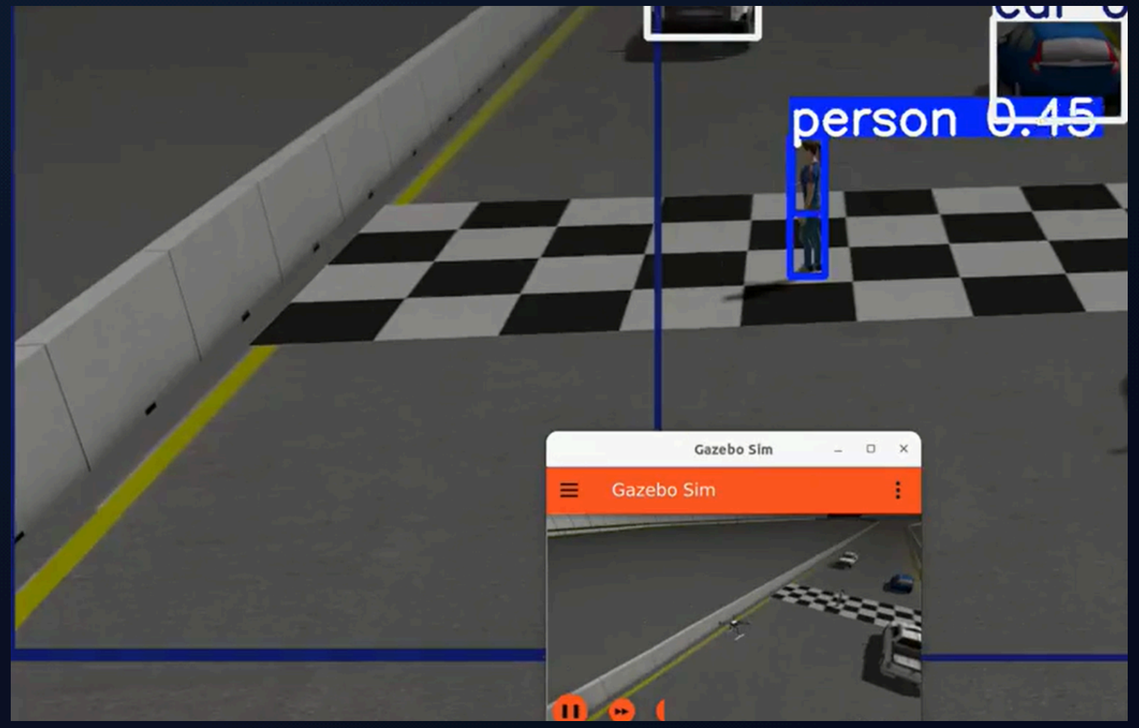
The Sonoma Raceway environment provides complex lighting and texture conditions, ideal for testing the robustness of the **YOLOv8** detection model.

Initial Challenges: Drone spawning "underground" and magnetic field interference due to local world properties.

```
PX4_GZ_MODEL_POSE="268.08,-128.22,3.86,0,0,-0.7"
```



Vision: Object Detection



Target Identification

We process the **IMX214** camera feed at 25-30 Hz to detect vehicles and people on the ground.

- **Model:** yolov8n (Nano) for performance.
- **Classes:** car, truck, bus, person.
- **Threshold:** confidence ≥ 0.40.
- **Filter:** Lock-on logic requiring 60 confirmed frames.

Coordinate Transformations

Mapping Frames

Critical for autonomous flight is the conversion between the **Inertial Frame (NED)** and the **Body Frame (FRD)**.

We use rotation matrices R_B to transform camera pixel coordinates to world offsets.

$$P_{world} = R_{drone} \cdot P_{body} + T_{drone}$$

PHASE 02

Kinematics & Dynamic Modeling

The mathematical core: modeling how motor speeds translate into 3D motion and trajectory tracking.

Direct Kinematic Model

The DKM maps the four individual rotor angular velocities to the total force and moments acting on the drone.

$$\begin{bmatrix} T \\ \tau_\phi \\ \tau_\theta \\ \tau_\psi \end{bmatrix} = \begin{bmatrix} c_f & c_f & c_f & c_f \\ 0 & lc_f & 0 & -lc_f \\ -lc_f & 0 & lc_f & 0 \\ c_m & -c_m & c_m & -c_m \end{bmatrix} \begin{bmatrix} \omega_1^2 \\ \omega_2^2 \\ \omega_3^2 \\ \omega_4^2 \end{bmatrix}$$

Quadcopter Mixing Matrix (X-Configuration)

Where c_f is thrust coefficient and c_m is moment coefficient.

Inverse Kinematic Model

IKM is used by the controller to calculate the required thrust and attitude to reach a desired velocity.

$$T_{req} = \frac{m\,(\,g + \ddot{z}\,)}{\cos\phi \cdot \cos\theta}$$

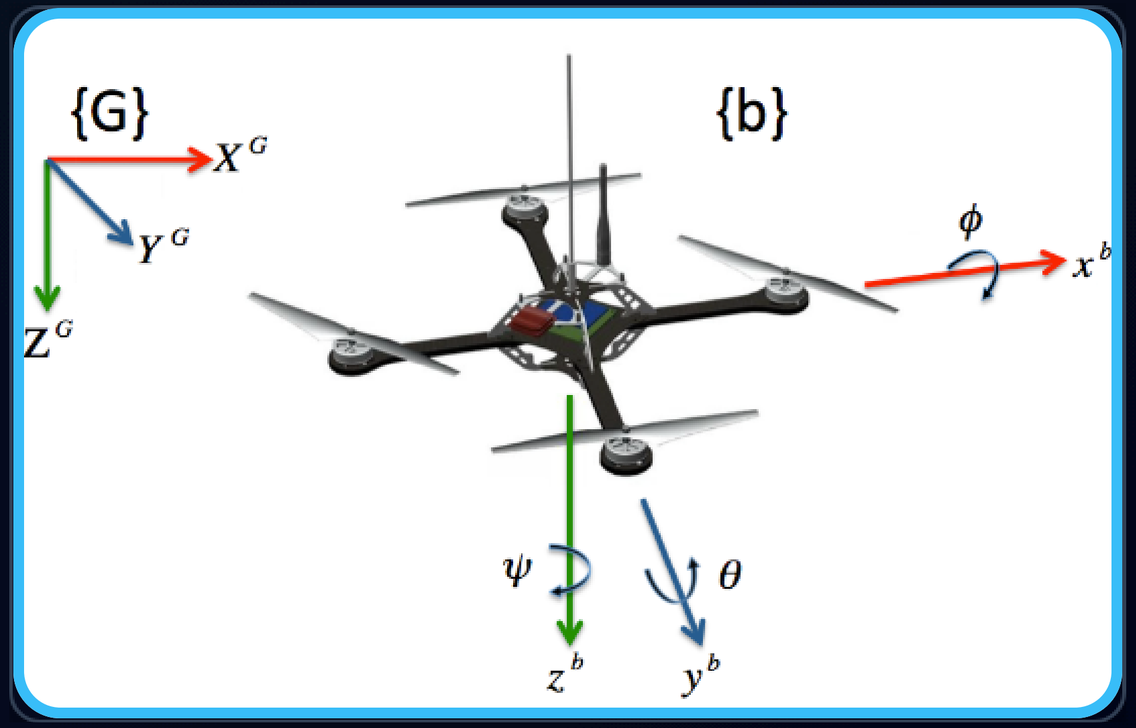
$$\phi = \tan^{-1}\left(\frac{\ddot{y}_{cmd}}{g + \ddot{z}_{cmd}}\right)$$

Translational Dynamics

Newton-Euler Equations

The center of mass acceleration is driven by gravity and the rotated thrust vector.

$$m\dot{v} = mge_3 - TRe_3$$



Rotational Dynamics

Modeling the angular acceleration of the quadrotor frame under torque and gyroscopic effects.

$$J\dot{\omega} = -\omega \times J\omega + \tau$$

Key Variables:

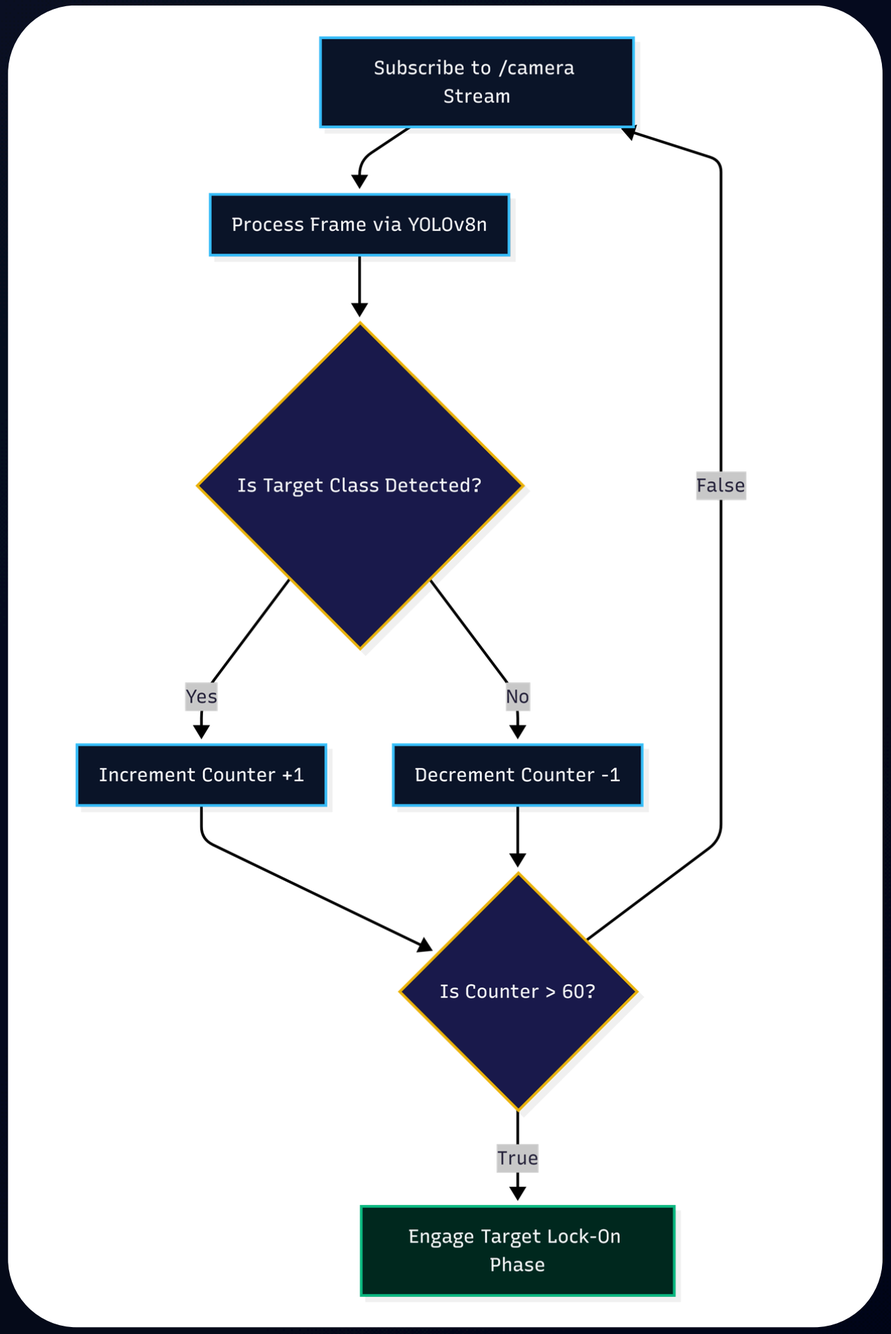
- J : Inertia Matrix (assumed diagonal).
- ω : Body angular velocity vector.
- τ : Resultant external torque vector.

Alg. 1: Detection Filter

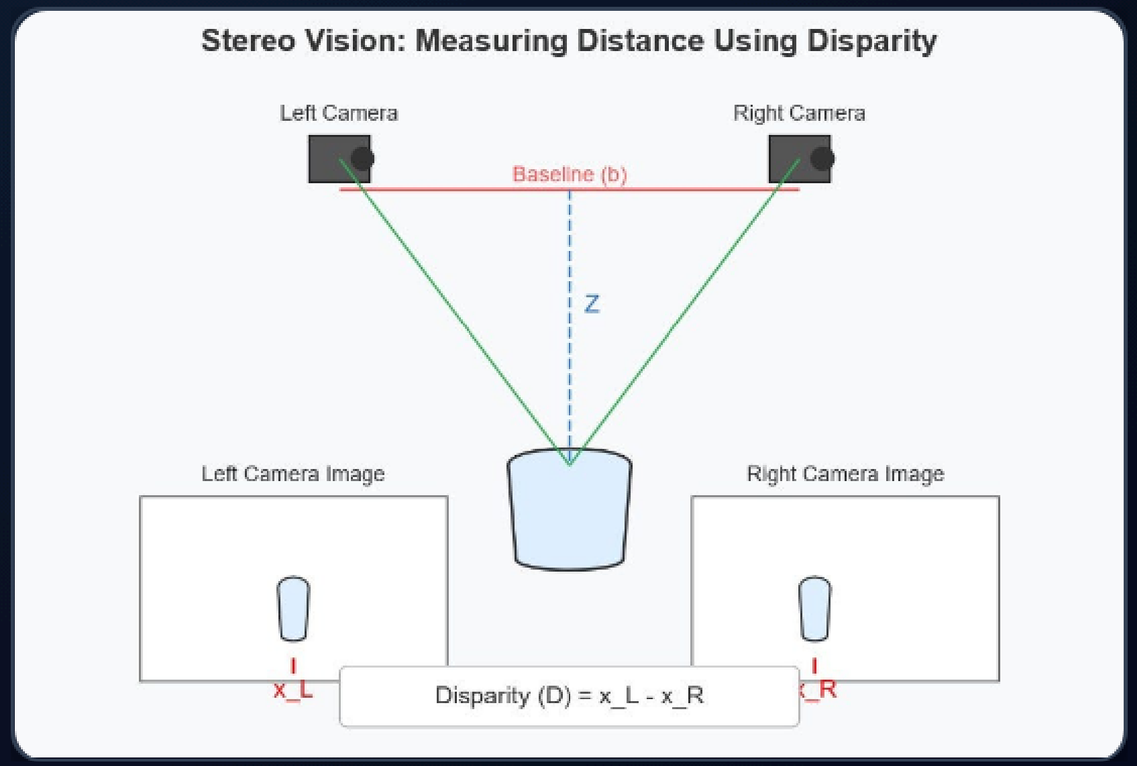
Detection Logic

We implement a "debounce" filter to prevent flight instability from false positives.

- 1. Subscribes to **/camera** topic.
- 2. Inference with **yolov8n.pt**.
- 3. Check if target class is vehicle.
- 4. **Counter:** If detection present, +1. If lost, -1.
- 5. **Trigger:** Lock target when count > 60.



Alg. 2: Pinhole Localization



Converts a 2D pixel coordinate into a relative 3D ground offset . (N, E)

$$E_{off} = \tan(\alpha_x) \cdot a$$

$$N_{off} = -\tan(\alpha_y) \cdot a$$

Where a is altitude and α_x is the horizontal angle off-axis.

Alg. 3: Offboard Guidance

Proportional Guidance

A capped proportional controller manages the approach speed to the ground target.

$$v_{cmd} = \frac{\min(v_{max}, dist)}{dist} \cdot e$$

Logic Flow:

- Distance $>$ threshold: Maintain **V_max**.
- Distance $<$ threshold: Ease in velocity (proportional deceleration).
- Distance $<$ arrival: Switch to **Position Hold** for delivery.

Engineering Challenges

The Problem	Our Solution
Black Camera Feed	Upgraded NVIDIA drivers & forced EGL vendor library to fix Gazebo rendering.
Magnetic Interference	Relaxed **EKF2_MAG_CHECK** in PX4 parameters to allow arming in custom worlds.
Drone "Coasting"	Implemented **Position Control** during final approach instead of pure velocity.
Detection Flicker	Introduced a temporal debounce filter requiring 60 consecutive frames.

Results & Analysis

95%

LOCK STABILITY

Flight Performance

The system demonstrated high reliability in detecting and tracking moving targets on the Sonoma Raceway racetrack.

- **Avg Inference:** 12ms (yolov8n).
- **Max Velocity:** 5.0 m/s during approach.
- **Stable Hover:** Verified ±0.1m accuracy.

Mission Execution Flow

Takeoff

Offboard climb to 6m altitude.

Search

Forward flight over racetrack world.

Target Locked

Visual confirmed approach phase.

Land/Deliver

Precise vertical touchdown & disarm.

Verified on PX4 v1.14 main branch & Gazebo Harmonic.

Thank You!

Questions & Live Demonstration

Autonomous Drone Surveillance Mission

Industrial Robotics Module 2026